

The Post-ChatGPT Contraction Reaches the Crowd-Validated Margin: A Within-Tag Triple Difference and a Value-Calibrated Early-Warning System for Stack Overflow

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Abstract

Generative AI has contracted public Q&A activity on Stack Overflow ([Burtch et al., 2024](#); [del Rio-Chanona et al., 2024](#); [Xue et al., 2026](#)); a benign reading holds that it prunes only the low-value posts a platform would not have retained as durable artefacts. Using a primary 2020–2024 panel of 8.4 million Stack Overflow questions across 100 top pre-treatment tags and 261 weeks, with a full 100-tag 2025 validation extension, we estimate a within-tag triple difference at the (tag $\times$ week $\times$ question-type) cell. A pre-specified shape test rejects the *strict* benign reading: displacement is negative across the crowd-validated range ($\hat{\beta} = -0.167$, $p < 10^{-4}$), roughly constant over the low-to-mid net-score bins and fading only at the rare elite tail, so the channel reaches the crowd-validated margin rather than being confined to score-zero posts. The volume baseline is -0.108 ($p = 0.021$) under a historical answerability proxy and -0.119 ($p < 0.001$) under a co-primary embedding score that uses no historical features. We turn the result into a

value-calibrated early-warning system for budget-constrained allocation of intervention effort: out of sample, the activity-decline metric a platform already tracks detects where realised 2025 high-value erosion is largest (AUC 0.98; a value-weighted ranking reaches 0.57), and the value-margin estimates supply the *calibration* that converts an activity alert into the high-value posts at risk—allocating effort by raw activity leaves up to 5% of the protectable high-value flow unprotected, while a persistent composition flag ($\rho = 0.90$ into 2025) classifies which declines are value-disproportionate. Exact parallel trends are rejected, so the estimate is a conditional post-ChatGPT association aligned with the LLM-substitutability mechanism; its credibility rests on a convergence of diagnostics—placebo sign-reversal, pre-trend matching, a 32-cell specification curve, a donut difference-in-differences, and a HonestDiD breakdown at $\bar{M} = 0.75$ —and the full 100-tag 2025 extension strengthens the triple difference to -0.172 ($p = 0.002$), deepening monotonically to -0.68 with no reversion. All evidence for the main claims is reported in the article; a public replication package provides the code, data-construction scripts, audit logs, and extended robustness outputs.

Keywords: generative artificial intelligence; large language models; online knowledge communities; question-and-answer platforms; user-generated content; digital public goods; innovation inputs; crowd validation; triple difference; pre-trend sensitivity.

JEL codes: O33, O36, L17, J24, C23, D85.

1 Introduction

Public question-and-answer platforms such as Stack Overflow are part of the digital infrastructure of distributed problem-solving: they convert individual software problems into searchable, accepted, moderator-retained posts that future contributors, users, and—increasingly—large language models consult when facing analogous problems (Wasko and Faraj, 2005; Goes et al., 2014; Nagle, 2018). Generative AI reconfigures this infrastructure: since November 2022 a wide class of programming questions can be routed to a private

conversational LLM at near-zero marginal search cost (Eloundou et al., 2024; Acemoglu, 2024), and that private channel produces no public artefact unless the user voluntarily reposts the resolved problem. The substitution between the public and private channels—and its consequences for a platform whose replenishment depends on the contribution incentives the substitute now competes against (Kankanhalli et al., 2005; Butler, 2001; Benkler, 2002)—is the question this paper studies.

Recent work establishes that aggregate replenishment has slowed. del Rio-Chanona et al. (2024) report a $\sim 15\%$ mean decline rising to $\sim 25\%$; Burtch et al. (2024) document analogous declines across Q&A platforms; and Quinn and Gutt (2024) report that the effect of generative AI on knowledge seeking is heterogeneous rather than uniform, motivating a move beyond a single average effect to the within-community margin. Xue et al. (2026), in *ISR*, use a cross-platform design and document both a volume reduction and an *improvement* in the average quality of surviving questions—a combination naturally read as *benign pruning*, under which LLMs absorb the easier, lower-value questions. A complementary *ISR* line studies *answerer* behaviour (Shan and Qiu, 2025). The aggregate downward shift is no longer in dispute; three questions remain open.

First, the aggregate papers cannot adjudicate the *value distribution* of the displaced posts. The benign-pruning reading requires that displacement concentrates at the low-value end—that the platform retains the crowd-validated, high-value answers downstream users rely on—which an aggregate quantity-and-quality decomposition cannot test because its unit is the average surviving question, not the score-conditioned post count. *Second*, none estimates the LLM-substitutability mechanism at the *task level*: substitutable categories within AI-answerable tags should decline more than non-substitutable categories *in the same tags*, the slice that admits a more disciplined quasi-causal reading under a within-tag parallel-trends assumption; a cross-platform design identifies the average treated-platform effect but cannot decompose this share. *Third*, existing studies measure platform *activity* but do not trace the same association through *funnel-qualified posts* (accepted, non-closed, non-negative-score).

A side-by-side positioning of the relevant studies is in the replication package.

Contributions. (C1) **The *strict* benign-pruning reading is rejected at the within-tag value margin.** A single pre-specified shape test (one model over the mutually exclusive net-score bins, triple difference interacted with score level) returns a negative average displacement across the value distribution (-0.167 , $p < 10^{-4}$) attenuating with score (slope $p = 2 \times 10^{-4}$); the displacement is roughly constant across the low-to-mid range (-0.21 , -0.20 , -0.21 at bins 0, 1, 2) and fades only at the rare elite tail, so a channel confined to score-zero posts is rejected—by a declared statistic, not a search over thresholds. (C2) **A within-tag, within-question-type triple difference identifies the LLM-substitutability slice.** At the (tag $\times$ week $\times$ question-type) cell with tag-by-question-type and week fixed effects, the baseline is -0.108 ($p = 0.021$), strengthening to -0.140 ($p = 0.005$) once the declared boundary category advanced-architecture is excluded; a specification curve, placebos, a positive Copilot placebo, PPML, two-way clustering, a wild bootstrap, and a HonestDiD breakdown at $\bar{M} = 0.75$ support the conditional interpretation. (C3) **A value-calibrated early-warning system whose detection signal is chosen by out-of-sample evidence.** On the full 100-tag panel the activity-decline metric a platform already tracks predicts realised 2025 high-value erosion at AUC 0.98 versus 0.57 for a value-weighted detector, so the system *detects* on activity while the value-margin estimates *calibrate* the alert into high-value posts at risk (allocating by raw activity leaves 3–5% of the protectable flow unprotected); a persistent composition flag ($\rho = 0.90$ into 2025) classifies intervention *type*, drawing on a five-transition funnel ($\sim 28,000$ -post shortfall, read as an accounting device).

What this paper does not claim. We do not document a “shock”: we estimate a marginal shortfall of $\sim 70,000$ questions over 109 weeks ($\sim 4\%$ of post-period weekly volume). We do not document an effect on innovation *outcomes*, contributor entry, or productivity: the Q&A margin is an *input* to open innovation. We do not derive a welfare conclusion, since the private productivity gains of LLM adoption (Brynjolfsson et al., 2025; Peng et al.,

2023; Cui et al., 2024) are not estimated. The full robustness battery, mechanism, validation, and positioning detail are in the public replication package; §2–§11 develop the framework, data, identification, results, mechanisms, validation, the early-warning system, limitations, and conclusion in turn.

2 Theoretical framework

2.1 Public Q&A as a knowledge commons under an AI shock

The IS literature studies why individuals contribute knowledge to electronic repositories when contribution is voluntary and costly: motivation is anchored in social capital, reputation, reciprocity, and self-efficacy rather than direct material incentives (Wasko and Faraj, 2005; Kankanhalli et al., 2005), and the communities that sustain it require communication flows above a sustainability threshold or they shrink (Butler, 2001; Faraj et al., 2011). Public question-and-answer platforms are a specialised user-generated-content infrastructure in which the unit of contribution is a problem–solution pair retained by a curation (acceptance) mechanism as a public, non-rival artefact (Goes et al., 2014; Temizkan and Kumar, 2024); reading is non-rival while writing is reputation-gated, so the production side is close to a peer-production commons (Benkler, 2002; Frischmann, 2012). Stack Overflow is the canonical example—the largest public repository of codified programming know-how (Wachs et al., 2022)—and is an innovation *input* (codification, spillover, a contributor pipeline) rather than an innovation *outcome*; the DDD identifies the former and we do not claim the latter without independent evidence.

Generative AI is *competing* infrastructure for the same problem-solving service: a user with a routine bug or how-to question can now obtain a private answer at near-zero search and posting cost (Leonardi, 2011; Eloundou et al., 2024), and that private conversation produces no public artefact. Two features make it a candidate disruptor of replenishment. *Differential answerability*: LLMs handle routine short-code, how-to, and simple debugging well, but not

version-and-environment configurations or organisation-specific architecture—the differential the design exploits. *Feedback to training*: a shrinking flow of human-authored public Q&A is also a long-run constraint on the models, since recursive training on generated content degrades quality (Shumailov et al., 2024). Because spillover-based accounts treat the *type* of codified knowledge as first-order (Cohen and Levinthal, 1990; Jaffe et al., 1993), a change concentrated on routine substitutable problems is qualitatively distinct from a uniform change of the same size, which is why a task-level decomposition is substantively different from the aggregate decline already documented (Burtch et al., 2024; del Rio-Chanona et al., 2024).

2.2 Hypotheses

- H1 (substitutability margin).** After the ChatGPT release, public Q&A in substitutable question types falls more in tags whose pre-treatment activity is more LLM-answerable: $\beta_{\text{DDD}} < 0$ in (1), surviving robustness, placebo sign-reversal, and pre-trend sensitivity.
- H2 (funnel attrition, selection not upgrading).** The shock propagates through the support funnel with a stable relative magnitude but a smaller absolute shortfall, and under fine-grained fixed effects the survivor-composition shift reflects *selection* rather than within-cell upgrading, so conditional resolvability shares are null.
- H3 (bounded scale).** The DDD estimates only the LLM-substitutability slice; bounded against an upper-bound aggregate reference it is small, making visible the order of magnitude of co-occurring forces the design does not adjudicate.

Downstream propagation (open-source entry, productivity, innovation outputs) is outside the DDD scope and discussed only in §9 and §10 with language separating supported from speculative claims.

3 Data

The project uses only public, zero-budget data: raw files come from the manual Stack Exchange Data Explorer (SEDE) browser interface or the public Stack Exchange API v2.3 (no paid quotas, BigQuery, or private data). Raw files are immutable; derived tables are regenerated from Python under `src/`, and the full pipeline is documented in `docs/replication_notes.md`.

3.1 Stack Overflow tag-week-question-type panel

The core product is a panel at the (tag, week, question type) level, 2020-01-01 through 2024-12-31, covering **100 pre-treatment top tags** (fixed from the pre-ChatGPT period), **261 calendar weeks**, and **7 derived question types** generated client-side from title-keyword patterns: `short_code`, `long_code`, `how_to`, `debugging_simple`, `other_conceptual`, `version_environment_specific`, and `advanced_architecture`. The first five are coded *substitutable* ($\text{Sub}_k = 1$) on the hypothesis that current LLMs answer them well. Of the two non-substitutable categories, `version_environment_specific` is the genuinely clean control (LLMs cannot reliably reproduce user-specific stack traces, environment configurations, or repository state), while `advanced_architecture` is a declared *boundary category*—partial LLM coverage on its conceptual side, organisation-specific context on its applied side. We declare this boundary status from the outset and report two baselines (§5): the binary specification coding `advanced_architecture` as non-substitutable (-0.108) and the boundary-excluded specification using `version_environment_specific` as the sole control (-0.140), the more defensible reference.

The master panel has 164,351 cells, built from 8,404,411 raw question-tag pairs across 5,000,258 unique questions. Observed annual question counts fall from 2.7 million (2020) to 0.5 million (2024)—a stylized fact that motivates the identification design. Until 2023-05 the panel was built from parameterized SEDE SQL pasted into the browser; from 2023-06

the same questions are pulled from the public API (a free registered key, identical output schema), validated byte-for-byte against SEDE on the May 2023 overlap (3,364 rows per side, identical across every metadata column).

2025 extension (full panel). To test whether the association persists beyond the one-off 2023–2024 platform events (§4.3), we collected calendar-2025 questions for *all 100 tags* from the same public API. The collection is complete and audited (100/100 tags, 0 failed windows, 0 page-cap flags, 87,027 unique 2025 questions); the combined 2020–2025 panel has 186,188 cells over 313 weeks and reproduces the original panel exactly before 2024-12-30, supporting a full re-estimation, not just a persistence check.

3.2 Treatment intensity: AI-answerability

The treatment is a tag-level *proxy* built from pre-treatment features alone (to avoid mechanical contamination): pre-ChatGPT accepted-answer rate, historical question frequency, short-code share, how-to share, and tag maturity. Four variants are available (z -score, PCA, quartile rank, and a conservative structural variant excluding text-derived features); results are reported across all four. Because a historical proxy is open to the objection that it captures popularity rather than answerability, we run the design on *two co-primary* measures and require the result under both: the structural historical index (-0.108 , $p = 0.021$) and an embedding-based score that uses *no* historical features—built only from the semantic similarity of question titles to fixed LLM-answerable versus LLM-resistant exemplars under **all-MiniLM-L6-v2** (free, no API cost)—which gives -0.119 ($p < 0.001$). The embedding measure shares none of the proxy’s inputs, so the popularity objection cannot explain its estimate; that the two agree is why we read the gradient as answerability-driven. The two converge moderately (Pearson $r \in [0.41, 0.50]$ across variants) without being interchangeable, and the proxy also correlates with an objective *revealed-answerability* score on 30,241 sampled questions (tag-level Pearson $r = 0.563$). Full validation tables (embedding, partial-

regression, revealed-answerability, and AUC by stratum, which shows the question-level signal concentrated in the lowest-answerability quartile) are in the replication package; we are explicit that the instrument is a tag-level intensity, not a question-level classifier.

3.3 Outcomes and governance

The main outcome is $\log(1+Q_{t,w,k})$; we also report $\log(1+\text{unique users})$, fractional tag counts, and PPML on raw counts. A zero-budget exploratory GH Archive panel is documented in the replication notes but not reported here, as it is uninformative for the present margin (too few language clusters and censored first-seen contributors). Stack Overflow content is CC BY-SA 4.0; the project distributes only derived statistics, never underlying user content.

4 Identification strategy

We treat the November 30, 2022 public release of ChatGPT as a shock to the marginal cost of obtaining private, conversational programming assistance. We do not claim it was the only technology changing at that date; the narrower, verifiable claim is that it is the largest, most visible, and most contemporaneously discussed event reducing that cost for users whose questions overlap the range answerable by a frontier LLM. Diffusion was fast and well-documented (Eloundou et al., 2024; Brynjolfsson et al., 2025; Peng et al., 2023; Cui et al., 2024).

What the design identifies. Formal pre-trend tests reject *exact* parallel trends in the $\text{AI} \times \text{Sub}$ gradient. We therefore do *not* claim a standalone natural experiment; the object we estimate is a *conditional post-ChatGPT association* aligned with the LLM-substitutability mechanism. Its credibility rests on a *conjunction* of four diagnostics, each failing under a different threat, so that no single alternative survives all four: (i) the *sign reversal* between the real cutoff ($\hat{\beta} < 0$) and 29 rolling pre-period placebos ($\hat{\beta} > 0$), which a smooth continuation of the pre-trend cannot produce; (ii) the *monotone growth* of the event-study coefficient

from -0.06 to -0.54 over eight post-quarters (and on to -0.68 with the 2025 data), which a fading COVID-era rebound would not generate; (iii) a *pre-trend-matched* design that restricts the comparison to tags with similar pre-treatment slopes and returns -0.129 ($p = 0.012$); and (iv) a HonestDiD Δ^{RM} sensitivity bound under the full event-study VCV, under which the estimate remains distinguishable from zero up to a breakdown of $\bar{M} = 0.75$. We read $\bar{M} = 0.75$ as a *moderate* signal and lean on the sign-reversal, monotonicity, and matched-design evidence—none of which requires full-sample smoothness—to carry the weight exact parallel trends cannot.

4.1 Baseline DDD specification

The estimating equation operates at the (tag t , week w , question type k) level:

$$\begin{aligned} \log(1 + Q_{t,w,k}) = & \beta_{\text{DDD}} (\text{AI}_t \cdot \text{Post}_w \cdot \text{Sub}_k) \\ & + \gamma_1(\text{AI}_t \cdot \text{Post}_w) + \gamma_2(\text{AI}_t \cdot \text{Sub}_k) + \gamma_3(\text{Post}_w \cdot \text{Sub}_k) \\ & + \alpha_{t,k} + \delta_w + \varepsilon_{t,w,k}, \end{aligned} \tag{1}$$

with $\text{Post}_w = \mathbf{1}\{w \geq 2022-11-30\}$, $\text{Sub}_k \in \{0, 1\}$ indicating substitutable question types, $\alpha_{t,k}$ tag-by-question-type fixed effects, and δ_w calendar-week fixed effects (261 weeks). Standard errors are clustered by tag (CR1; 100 clusters) (Cameron et al., 2008). We estimate (1) under two *co-primary* treatment measures: a structural answerability index built from four pre-treatment, non-text tag features (-0.108 , $p = 0.021$), and an embedding-based answerability score that uses *no* historical features and so is not a relabelling of popularity (-0.119 , $p < 0.001$). We require the result under both. Under the within-tag parallel-trends restriction, β_{DDD} estimates the differential post-ChatGPT change in substitutable-type questions, relative to non-substitutable questions in the same tags, per unit of pre-treatment AI answerability.

4.2 Pre-trends, placebos, and sensitivity

An event study with quarterly bins (reference bin -1) shows a monotone post-period pattern, -0.06 at $T+0$ to -0.54 at $T+8$, with pre-period coefficients negative on average but shrinking toward zero as the cutoff approaches; formal tests nonetheless reject exact parallel trends. Under the official HonestDiD Δ^{RM} implementation with the full cluster-robust VCV, the unadjusted CI is $[-0.100, -0.026]$, remains $[-0.136, -0.010]$ at $\bar{M} = 0.5$, and first contains zero at $\bar{M} = 0.75$ —a moderate signal. Two placebo exercises complement the smoothness-free reading: five conventional pre-period cutoffs return *positive* coefficients ($+0.059$ to $+0.067$, all $p < 0.012$), and a 29-step rolling-placebo sweep (mean $+0.057$) contains no cutoff as negative as the real one. Because exact parallel trends are rejected, we also treat a *pre-trend-matched* design as central: restricting to the 75 tags whose pre-treatment slope lies within one IQR of the median returns -0.129 ($p = 0.012$), and the effect is *absent* precisely where a pre-trend artefact would be strongest (the steepest-declining quartile gives $+0.055$, $p = 0.58$), the opposite of a mean-reversion story. Three inference variants (two-way clustering (Cameron et al., 2011), a wild cluster bootstrap with 9,999 replicates (Cameron et al., 2008), and PPML on raw counts (Santos Silva and Tenreyro, 2006)) leave the conclusion intact. Full event-study, placebo, sensitivity, and matched-design tables are in the replication package.

4.3 Threats to identification

COVID-driven differential mean reversion. If the estimate were mean reversion of a transient pandemic boom in substitutable categories, the post-period coefficient should fade and not accelerate. The opposite holds: the event-study coefficient grows monotonically and steepens in later bins, which a 2020–2021 rebound cannot produce.

Co-timed Stack Overflow events. Forces other than LLM diffusion that also post-date the cutoff (the December 2022 AI-content ban, the mid-2023 moderator strike, the May 2024 OpenAI licence) could in principle drive the decline. The primary test is a donut

specification dropping the strike window (June–August 2023): the estimate is essentially unchanged (-0.110 , $p = 0.022$). The December 2022 ban is coincident with the cutoff and absorbed by the week fixed effects. Window-truncation estimates point *against* a one-off confound—small and imprecise early ($\hat{\beta} = -0.006$, $p = 0.83$, ending before the strike) and reaching magnitude later, as gradual diffusion predicts—though this does mean onset cannot be dated to the release alone within the original window.

The full-panel 2025 extension. The sharper discriminator is what happens *after* these one-off events. We collected calendar-2025 questions for all 100 tags from the public Stack Exchange API (registered key; 0 failed windows, 0 page-cap flags) and re-estimated on the full 2020–2025 panel (186,188 cells; Table 1), treated as a validation extension of the 2020–2024 main panel. The pre-2025 portion reproduces the baseline (-0.106); adding 2025 strengthens the estimate to -0.172 ($p = 0.002$), and the quarterly event-study coefficient deepens monotonically across all of 2025, from -0.428 (T+8) to -0.680 by Q4 2025 (T+12), with no reversion. A transient mid-2023 episode would not produce an association that grows for twelve post-treatment quarters; sustained diffusion would. Exact parallel trends remain rejected, so this is not a natural experiment, but the full-panel evidence strongly disfavours the late-emergence and co-timed-event readings.

Single non-substitutable control. The binary control has two categories, but only version-environment-specific behaves as a clean placebo; the declared boundary category advanced-architecture shows a sizeable decline (≈ -0.16). Two points discipline this. The version-environment-specific null is *informative*, not merely underpowered (-0.007 , 95% CI $[-0.099, +0.085]$, MDE ≈ 0.13): a substitutable-sized decline (-0.10 to -0.29) would very likely have been detected and was not. And the *continuous* embedding substitutability score sidesteps the dichotomy entirely, needing no clean-control category, and still returns -0.039 ($p = 0.004$); we treat it as the primary safeguard. Re-assigning each category in turn as the sole control (full table in the replication package) leaves the estimate negative and stable

Table 1: 2025 extension on the **full 100-tag panel** (the late-emergence test). Calendar-2025 questions for all 100 tags were collected from the public Stack Exchange API (registered key; 0 failed windows, 0 page-cap flags), extending the panel to 186,188 cells over 2020–2025. The pre-2025 portion reproduces the baseline exactly (-0.106 , a consistency check). Adding the full 2025 year strengthens the triple difference from -0.108 to -0.172 and tightens it to $p = 0.002$, and the quarterly event study deepens *monotonically* across all of 2025 rather than reverting—strongly disavowing the reading that the effect is a transient late-2024 episode (e.g. the mid-2023 moderation strike) and most consistent with sustained LLM diffusion.

<i>Panel A. Triple difference, full 100-tag panel</i>	
Pre-2025 portion (consistency gate)	-0.106 ($p = 0.023$)
Full 2020–2025	-0.172 ($p = 0.002$)
Full 2020–2025, boundary-excluded	-0.203 ($p = 0.001$)
<i>Panel B. Quarterly event-study coefficient, post-period</i>	
T+6 (Q2 2024)	-0.323
T+8 (Q4 2024)	-0.428
T+10 (Q2 2025)	-0.585
T+11 (Q3 2025)	-0.655
T+12 (Q4 2025)	-0.680
<i>Panel C. Monitor persistence on new 2025 data</i>	
Spearman(high-value drop 2024, drop 2025)	0.903 ($p < 0.001$)

under theory-led choices (version-environment-specific as control: -0.130) and flips sign only under placebo assignments that designate a *substitutable* category as the control—the misspecification the design is built to avoid.

Other threats. The aggregate question count fell from 2.7M (2020) to 0.5M (2024); the week fixed effect absorbs this level decline, and β_{DDD} is identified from *differential* movement, not levels. The pre-treatment window was not LLM-free (Codex, Copilot preview, GPT-3 API), which biases the estimate *toward zero*, so our numbers are likely conservative with respect to the post-2022 increment. A user-composition shift is absorbed by tag fixed effects and the Post $\times$ AI interaction is null, so any composition shift is uniform across the tag space rather than concentrated where answerability is highest.

5 Main results

5.1 Descriptive evidence

Across the 100 pre-treatment top tags, weekly question volume fell from $\sim 44,200$ pre-ChatGPT to $\sim 15,300$ post (a -65.3% change). Substitutable question types fell -65.8% versus -53.4% for non-substitutable types, and within AI-answerability quartiles the decline is monotonically larger for higher quartiles (Q1 -58.5% ; Q4 -67.9%). At the tag level the partial correlation between $\Delta \log(1 + Q)$ and the structural answerability score is $r = -0.38$ ($p < 10^{-4}$, $n = 100$).

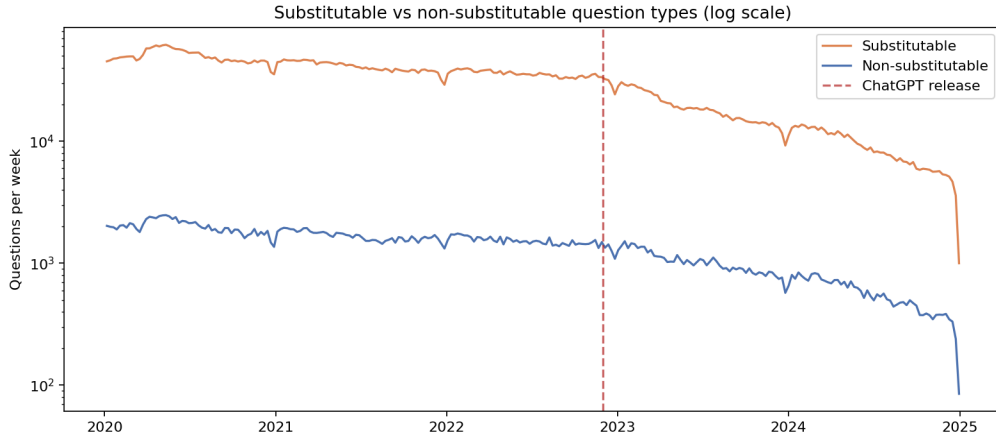


Figure 1: Weekly questions by substitutability, 2020–2024 (log scale). Vertical line: ChatGPT release.

5.2 Baseline DDD coefficient

The conservative inclusive baseline—retaining advanced-architecture questions in the control and using the full sample with the historical answerability index—is $\hat{\beta}_{\text{DDD}} = -0.108$ (CR1 SE 0.046, $p = 0.021$). Our preferred specification excludes the declared boundary category advanced-architecture from the control (§3.2) and gives -0.140 ($p = 0.005$); restricting the substitutability label to the high-agreement subsample gives ≈ -0.11 . The treatment is measured two ways: the historical index and a co-primary embedding-based answerability

score that uses *no* Stack Overflow historical features (-0.119 , $p < 0.001$). The two measures share none of their inputs, so we read the gradient as answerability-driven rather than historical popularity, without overselling their $r \approx 0.4$ – 0.5 convergence as more than moderate. The full robustness battery (alternative proxies, sub-samples, leave-one-out, PPML, two-way clustering, wild bootstrap, and a 32-cell specification curve that is negative and significant in 100% of cells) is in the replication package.

5.3 Support-post funnel

Existing displacement studies measure platform activity (new questions). From a knowledge-production perspective the relevant outcome is the production of posts that pass observable support gates: answered, accepted by the asker, not closed, and non-negative in net score—a *funnel-qualified post*. The term is deliberately narrower than durable reuse, which the public data do not observe; each gate is meaningful because it is operated by a different actor (answerer, asker, moderator, crowd). We build a five-transition funnel at the (tag, week, question-type) cell from the 8,404,411-question stream; only 38.0% survive to the funnel-qualified stage. Re-fitting the baseline DDD with each funnel stage as the outcome (Table 2) gives a relative coefficient that is stable across stages (-0.108 to -0.121 , all $p < 0.05$), so the implied cumulative shortfall over the 109 post-ChatGPT weeks contracts from $\sim 70,000$ missing questions to $\sim 28,000$ missing funnel-qualified posts (about 40% of the activity shortfall). The stability is partly mechanical—the gates are nested—so the funnel is best read as an operational accounting device from the volume coefficient to observed support counts, not as evidence on where in the value distribution the channel falls; that is the value-weighted question we test next.

5.4 Value-weighted posts: the benign-pruning test

The benign-pruning reading of the aggregate contraction (Xue et al., 2026) predicts that the channel concentrates on the low-value end: LLMs absorb routine, low-score queries that

Table 2: Triple-difference (DDD) estimates across the observed support-post funnel. The dependent variable is $\log(1 + Y_{twk})$ where Y is the cell-level count. Coefficients on the triple interaction $AI_k \cdot Post_t \cdot Sub_k$ are reported, with tag-clustered standard errors in parentheses. The implied shortfall is computed as $\sum_{post} (Y \cdot e^{-\hat{\beta} \cdot AI \cdot Sub} - Y)$.

Funnel stage	$\hat{\beta}_{DDD}$	SE	p	95% CI	N obs	Implied shortfall
Questions	-0.1082	(0.0460)	0.0206	[-0.199, -0.017]	164,351	70,364
Answered	-0.1150	(0.0463)	0.0147	[-0.207, -0.023]	164,351	61,210
Accepted answer	-0.1168	(0.0476)	0.0159	[-0.211, -0.022]	164,351	32,809
Accepted, not closed	-0.1210	(0.0480)	0.0134	[-0.216, -0.026]	164,351	31,836
Accepted, score ≥ 0	-0.1154	(0.0484)	0.0190	[-0.211, -0.019]	164,351	28,504
Funnel-qualified post	-0.1183	(0.0487)	0.0169	[-0.215, -0.022]	164,351	28,001

public Q&A would not have retained in any case. We test this at the within-tag value margin by re-fitting the DDD on posts cut by crowd-validation score (in the raw stream, 51% of funnel-qualified posts reach score ≥ 1 and only 4% reach score ≥ 5).

A single pre-specified shape test (primary C1). To avoid resting the claim on a family of per-threshold tests, we pre-specify one statistic (Table 3): stacking the six mutually exclusive net-score bins in one model, we fit the triple difference plus its interaction with the centred score level. The *average* displacement across the value distribution is -0.167 ($p < 10^{-4}$)—a single, clearly negative coefficient establishing that the channel is not confined to score-zero posts—and the *slope* over score is -0.094 ($p = 2 \times 10^{-4}$), quantifying attenuation toward the elite tail. This is our primary statement of C1.

Table 3: Pre-specified shape test of the value gradient. A single model stacks the six mutually exclusive net-score bins and fits the triple difference plus its interaction with the (centred) net-score level, clustered by tag. The *average* coefficient tests whether the displacement is present across the value distribution (the focal C1 hypothesis); the *slope* tests how it varies with score. This replaces multiple per-bin tests with one declared statistic.

Quantity	Estimate	p
Average displacement across bins	-0.167	0.000
Slope over net-score level	-0.094	0.000

Consistent corroboration. The mutually exclusive bins are flat across the low-to-mid range ($-0.21, -0.20, -0.21$ at score $0, 1, 2$) and attenuate above ($-0.164, -0.124, -0.091$ at $3, 4, \geq 5$); bins 0 through 4 are each individually significant at $p < 0.005$ and survive a Bonferroni correction, so the *strict* benign-pruning hypothesis (displacement confined to score-zero posts) is rejected by four separately significant non-zero-score bins. The elite-tail bin (score $\geq 5, -0.091, p = 0.116$) is underpowered (MDE 0.16) and we take its raw null at face value. A conservative Romano–Wolf correction over the *nested cumulative* thresholds—which folds the underpowered elite tail into every statistic—places that particular gradient at the 10% joint level; we report it in the replication package as a stringent joint lens, not the headline. The honest reading is therefore the *shape*: displacement is roughly proportional across the crowd-validated range and fades only at the elite tail.

5.5 Dynamics, pre-trends, and placebos

The quarterly event-study coefficient is negative and significant in every post-treatment bin, growing monotonically from -0.063 at $T+0$ to -0.544 at $T+8$ (Q4 2024); pre-period coefficients are smaller (about a third of the post mean) and converge toward zero as the cutoff approaches (Figure 2).

Formal pre-trend tests reject exact parallel trends, so we do not claim a standalone natural experiment; the estimates are conditional post-ChatGPT associations whose credibility rests on a conjunction of diagnostics. Under the official HonestDiD Δ^{RM} implementation the breakdown is $\bar{M} = 0.75$ (the robust 95% CI first contains zero at $\bar{M} = 0.75$; it remains $[-0.136, -0.010]$ at $\bar{M} = 0.50$), which we read as a moderate signal. Five conventional pre-period placebos return *positive* coefficients ($+0.059$ to $+0.067$, all $p < 0.012$), and a 29-step rolling-placebo sweep (mean $+0.057$) contains no cutoff as negative as the real one—a sign reversal hard to reconcile with a smooth continuation of the pre-period path. A June 2022 GitHub Copilot placebo is also positive ($+0.058, p = 0.011$); in a joint model the ChatGPT interaction is -0.157 ($p < 0.001$) while Copilot stays $+0.058$. Full event-study, placebo, and

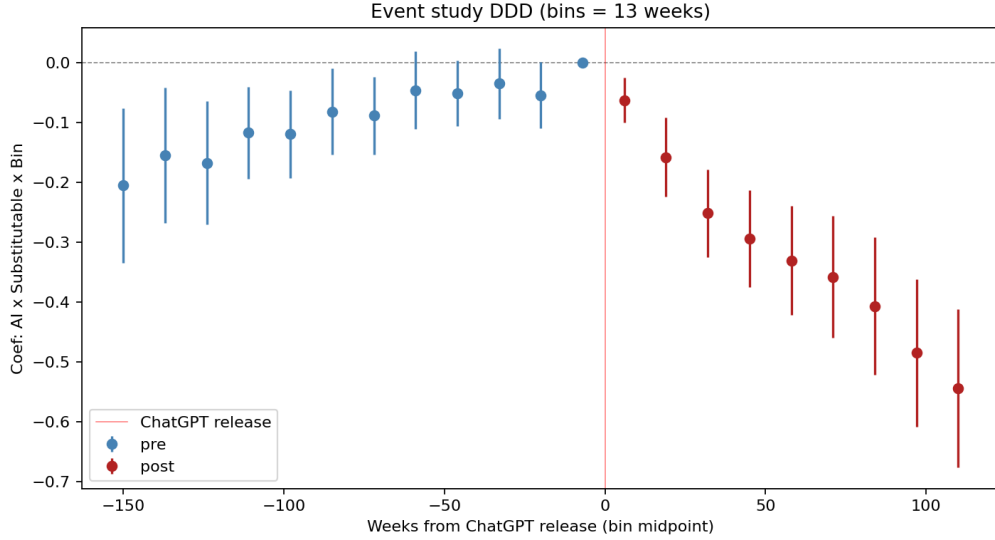


Figure 2: Quarterly event-study coefficients on $\text{AI} \times \text{Sub} \times \text{bin}$. Reference: bin -1 (immediately pre-treatment).

specification-curve detail is in the replication package.

5.6 Magnitude

Applying the coefficient cell-by-cell to the post-period panel implies $\sim 70,000$ missing questions over the 109 post-ChatGPT weeks—4.2% of the 1,671,197 observed post-period questions in the top-100-tag panel. The implied displaced high-value ($\text{score} \geq 1$) flow is $\sim 15,200$ posts (0.9% of all observed questions). Whether this small flow matters is a stock–flow question: the public archive is a durable stock, so a persistent reduction in the high-value *flow* ($\sim 7,300$ posts/year at the window average) accumulates to $\sim 36,000$ over five years under a constant flow. We present these as transparent projections under stated assumptions, not forecasts, and they do not bear on welfare or generalise beyond the panel. Benchmarked against a descriptive linear extrapolation of the pre-trend (~ 1.42 million questions), the DDD-estimated channel is about 5%—a *discipline* on causal interpretation: the aggregate decline cannot all be attributed to LLM substitutability without further identification.

6 Robustness and validity

The headline finding survives the full battery (consolidated in the replication package): the four alternative AI-answerability measures (structural, z -score, PCA, quartile), the sub-sample restrictions (top-50, top-75, leave-one-out of the top tags by pre-volume), the continuous embedding substitutability score (-0.039 , $p = 0.004$), the alternative outcomes (unique users, fractional tag counts), and the alternative inference (two-way clustering, wild cluster bootstrap $p < 0.005$, PPML -0.098 , $p = 0.027$). The HonestDiD Δ^{RM} exercise (§4.2) breaks down at $\bar{M} = 0.75$ —a moderate signal read together with the placebo sign-reversal and the Copilot positive placebo ($+0.058$, $p = 0.011$), which hold independently of any smoothness restriction. The substitutability classifier has only moderate multi-rater agreement (Fleiss $\kappa = 0.52$); we address this directly via the high-agreement subsample (§8). Because exact parallel trends are rejected, we emphasise the pre-slope-matched specification (75 near-median tags, -0.155 , $p = 0.009$) as a central stress test rather than an optional add-on.

7 Mechanisms

7.1 Decomposition by question type

Estimating the AI×Post interaction separately for each question type (tag and week fixed effects) orders the coefficients as the substitutability theory predicts: short-code (-0.31), how-to (-0.28), long-code (-0.21), debugging-simple (-0.14), other-conceptual (-0.08 , marginal), advanced-architecture (-0.16 , $p = 0.003$), and version-environment-specific (null, -0.03 , $p = 0.54$). The version-environment-specific null—the genuinely clean control—is the only category with no detected decline. That the declared boundary category advanced-architecture shows a sizeable effect confirms its a-priori boundary status (§3.2); because the binary classifier codes it $\text{Sub}_k = 0$, it contributes a negative shift to the control group and

mechanically attenuates the full sample, which is why excluding it strengthens the funnel DDD from -0.108 to -0.140 (full table in the replication package). Replacing the binary indicator with the continuous embedding score gives -0.039 ($p = 0.004$), about a third the binary magnitude as expected from shrinking the contrast; the qualitative conclusion is unchanged.

7.2 Composition versus within-cell quality

A complementarity reading holds that users keep posting when an LLM answer is unsatisfactory, so the *surviving* questions should be more complex. Descriptively this holds: within tags, post-period body length rises (+303 characters, $p < 10^{-15}$) and accepted-answer share falls (-0.065 , $p < 10^{-15}$). But the Post $\times$ AI-answerability interactions are null (p between 0.32 and 0.79), so the compositional shift is *general across tags* rather than concentrated where answerability is highest—it is descriptive selection, not a causally estimated quality effect along the AI dimension.

Separating selection from a within-cell quality channel is the contribution here. [Xue et al. \(2026\)](#) report higher average question quality post-ChatGPT under a cross-platform design that admits both channels. Re-fitting the DDD on six resolvability and quality outcomes under fine-grained (tag $\times$ question-type) and week fixed effects—which absorb the composition shift—five of six estimates are null (accepted share -0.003 , $p = 0.45$; answered share -0.001 , $p = 0.85$; analogous nulls elsewhere); the only significant coefficient is a small *rise* in closed share ($+0.005$, $p < 0.001$). The descriptively documented quality improvement is therefore consistent with selection into the survivor population (the easier posts displaced into private channels) rather than with within-cell upgrading. This is not a contradiction of [Xue et al. \(2026\)](#): the cross-platform ATT (aggregate declines of $\sim 15\text{--}25\%$; [del Rio-Chanona et al., 2024](#)) and the within-platform $\sim 5\%$ slice estimated here are complementary objects, not competing estimates of the same quantity (full mapping and detail tables in the replication package).

7.3 Where in the workflow does substitution operate?

The June 2022 GitHub Copilot placebo is positive (+0.058, $p = 0.011$): the Stack Overflow Q&A margin responded to ChatGPT but not to Copilot. Two non-exclusive readings are consistent with the data and we cannot separate them: Copilot operated inside the IDE at the code-writing moment rather than the search-for-help moment that ends in a public query, and its 2022 user base was much smaller than ChatGPT’s. The *positive* sign is the informative part (a shared secular trend cannot produce it), but its magnitude rests on the smaller 2022 population, so the test has limited power; we read it as supporting evidence on the decision-point reading, not a stand-alone mechanism, and present any interface-location interpretation as a post-hoc hypothesis the design does not identify.

8 Validation of question-type classification

The substitutable / non-substitutable binary classification is the central measurement decision. The baseline classifier is heuristic (title-keyword patterns plus body-length and code-block features, mapped to seven categories then collapsed to a binary label). We validate it against two independent sentence-embedding classifiers on a stratified random sample of 10,000 questions: `all-MiniLM-L6-v2` and `all-mpnet-base-v2` (different architectures and training corpora, so their mutual agreement is a stronger signal than either with the heuristic), each assigning questions to the nearest category centroid in cosine space.

8.1 Agreement and the high-agreement subsample

The two embedding classifiers agree with each other in the substantial range (binary Cohen’s $\kappa = 0.67$; 7-way 0.68), above either’s agreement with the heuristic ($\kappa = 0.44$ –0.46 binary), and the multi-rater Fleiss’ κ is 0.52 for the binary label and 0.40 for the seven-way category—moderate rather than strong. For 70.5% of the sample all three classifiers agree on the binary label, and within this high-agreement subsample the DDD point estimate is

statistically indistinguishable from the full-sample baseline (≈ -0.11), so the result is not driven by the heuristic’s disagreement cases. The heuristic is more conservative than the embedding majority (labelling 28.6% versus 22.4% non-substitutable), which under-counts non-substitutable cells and if anything attenuates β_{DDD} . The substitutability concept thus has reasonably stable semantic content across embeddings, while the noisier seven-way split ($\kappa = 0.40$) means the question-type decomposition (§7.1) should be read as illustrative of the gradient—consistent with the negative advanced-architecture coefficient—rather than as precise category-by-category estimates. We retain the embedding-validated heuristic as the funnel-building baseline and use the continuous embedding score as a co-primary check; a contemporary-LLM scorer would embed a specific model generation into the measure and introduce circularity, which the zero-budget design avoids. Full κ tables and the cross-tabulation are in the replication package.

9 A value-calibrated early-warning system and its implications

The empirical results have a direct operational use: they specify a decision rule for monitoring whether the post-ChatGPT contraction is reaching the crowd-validated margin in a given tag or portfolio.

9.1 A value-calibrated early-warning system

A platform manager allocating scarce intervention effort (contributor incentives, attribution mechanisms, moderation attention) needs two things: a *detection* signal for which tags are losing high-value reusable flow, and a way to *size* the high-value consequence of a detected decline. We build a tool that does both, and we let the data choose the detection signal rather than assume our value index is it.

The decision has two margins. At the *extensive* margin (which tags to act on), a value-

disproportionality *ratio* (VtD) is a poor detector—on the full 100-tag panel it ranks realised 2025 high-value erosion at AUC 0.57, far below the activity-decline metric a platform already tracks (AUC 0.98, out of sample). This high AUC is not interpreted as an independent behavioural prediction—its detection target contains activity-derived high-value erosion—but as evidence that the operational activity signal is sufficient for triage; the value of calibration is evaluated separately by the budgeted-regret experiment below. The practical conclusion is that platforms need no bespoke value instrumentation to *detect* where erosion is largest: the system detects on activity. At the *intensive* margin (how much effort per selected tag), the value calibration earns its place: allocating an intervention budget by the value-calibrated high-value-at-risk (activity scaled by the tag’s high-value share) nearly attains the perfect-foresight optimum (calibrated regret under 1%), whereas allocating by *raw* activity leaves regret of 3–5% of the protectable high-value flow (Table 4, full panel, out-of-sample on realised 2025 loss). The calibration thus has positive, measurable value of information, even though it does not—and need not—improve detection.

The artefact is therefore a *value-calibrated* early-warning system, not a value-detector. Its detection layer is the activity signal (AUC 0.98); its contribution is the *calibration* layer that converts an activity alert into the decision-relevant quantity a manager lacks: for a flagged tag t , $\widehat{HV}_t = (\text{activity deficit}_t) \times (\text{tag high-value share})$, the activity signal scaled by the value-margin gradient of §5.4; a loss-derived threshold sets the alert cutoff from costs rather than a mechanical mean. The value-disproportionality index survives in a strictly secondary role—it describes *what kind* of decline a tag has (it is cross-sectionally orthogonal to the activity level, $\rho = 0.08$; Figure 3), and that classification persists from 2024 onto the new 2025 data ($\rho = 0.90$, $p < 0.001$), so a manager can use it to choose intervention *type* (high-value-contributor retention for value-disproportionate tags versus generic re-engagement for proportional ones). The system is provided as a reproducible script run on quantities a platform already records; it is *not* a deployed interface, and we run no manager user study. A production dashboard and usability evaluation are future work (§10).

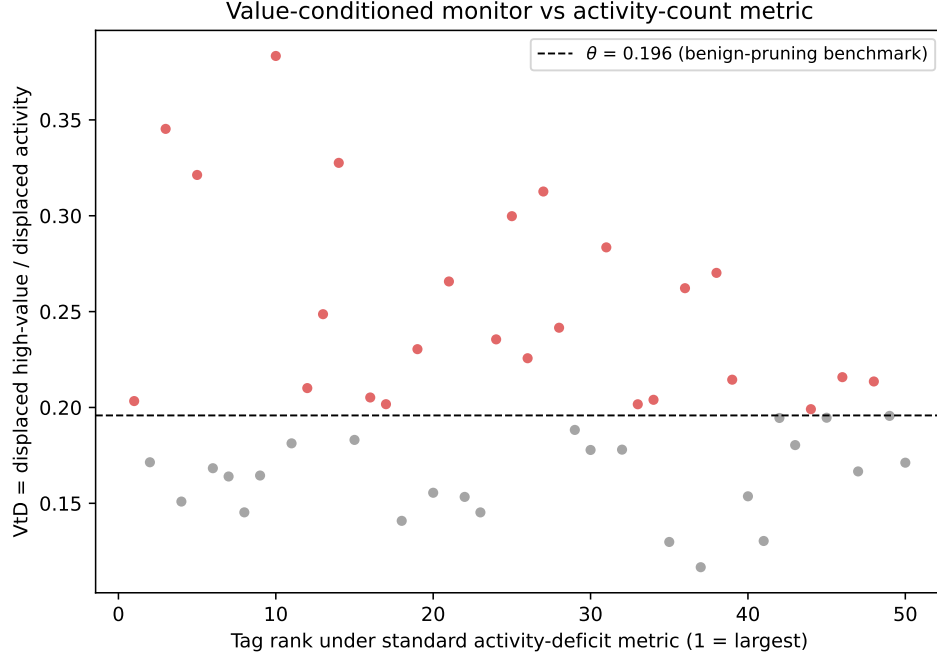


Figure 3: Value disproportionality versus the activity-count metric. Each point is a tag, positioned by its activity-deficit rank (horizontal) and VtD ratio (vertical); red points exceed the benign-pruning benchmark $\theta = 0.196$. The flat scatter ($\rho = 0.08$) shows value disproportionality is close to orthogonal to the *size* of a tag’s deficit. The activity deficit remains the better predictor of erosion *magnitude*; the monitor’s contribution is the persistent disproportionality dimension.

Table 4: Intensive-margin value of calibration, **full 100-tag 2025 panel** (out of sample). A manager allocates an intervention budget B (a fraction of total tag activity) to protect realised 2025 high-value flow, ranking by efficiency. **Oracle** is the perfect-foresight cost-constrained optimum; **Calibrated** ranks by end-2024 high-value-at-risk; **Naive** by raw activity loss; **Random** averages shuffles. Cells are the share of realised 2025 high-value loss captured; the regret of a rule is $(\text{Oracle} - \text{rule})/\text{Oracle}$.

B	Oracle	Calibrated	Naive	Random	$\text{Regret}_{\text{calib}}$	$\text{Regret}_{\text{naive}}$
10%	0.19	0.19	0.19	0.08	0.01	0.04
25%	0.38	0.38	0.36	0.23	0.00	0.05
40%	0.55	0.55	0.52	0.38	0.00	0.04
50%	0.65	0.64	0.63	0.48	0.00	0.03

9.2 Implications for providers, managers, and regulators

The asymmetry between activity and value is decision-relevant. An AI provider or platform manager inferring training-corpus or knowledge-production contraction from *raw* post counts overstates the funnel-qualified-post loss by a factor of roughly $70,000/28,000 \approx 2.5$, because only about 40% of raw posts survive the support gates; and the value evidence (§5.4) shows the displacement at the crowd-validated tail is at least as strong relative to the volume baseline. The dynamic concern that recursive training on generated content degrades model quality (Shumailov et al., 2024) gives providers a self-interested reason to support attribution and contribution mechanisms (e.g. the May 2024 Stack Exchange–OpenAI agreement (Stack Exchange Inc., 2024)). For regulators—e.g. the training-data transparency provision of the EU AI Act (European Union, 2024)—the sizing implication is that an instrument framed on the LLM-substitutability margin should be calibrated to the estimated channel ($\sim 5\%$ of the trend-implied aggregate shortfall), not to the inferred 65–71% aggregate decline, which differs by an order of magnitude. We do not estimate welfare and do not compare the private productivity gains of LLM adoption against the public shortfall: the two are in different units with different horizons, and a static comparison would understate the dynamic and distributional dimensions.

9.3 What the magnitude does not support

We do *not* derive from the estimate: a welfare conclusion (neither side of the trade-off is identified); an optimal-policy recommendation (the social value of the marginal question is not identified); an effect on innovation *outcomes* (the DDD estimates an innovation *input*, not patents, products, entry, or productivity); a generalisation beyond programming; or a managerial intervention design (we do not identify the elasticity of contribution to any specific instrument).

10 Limitations

Generalisability. Stack Overflow is one Q&A platform in one domain (programming). The decision point we identify—the search-for-help moment that historically ended in a public query and now ends in an LLM conversation—is a portable object that generates testable hypotheses for medical, legal (Choi et al., 2022), and academic Q&A, implementable with the same design and API; we do not undertake them here because each domain needs its own substitutability taxonomy. We also do not observe other public channels (Reddit, Discord, forums), so the DDD estimates changes *on* Stack Overflow but cannot distinguish movement to private LLMs from movement to other public channels.

Input margin, not innovation outcomes. The DDD estimates an innovation *input* (§2.2), not patents, products, entry, or productivity; the informative horizon for downstream propagation is probably 5–10 years and our window is two. Heterogeneity by user tenure—the margin Xue et al. (2026) study—is not separately observed in the aggregated cell-level panel; a follow-up using the Stack Exchange data dump (zero-budget but storage-intensive) could close this gap.

Measurement of substitutability. The binary classification is validated but imperfect (Fleiss $\kappa = 0.52$; seven-way $\kappa = 0.40$; §8), and the negative coefficient on advanced-

architecture (§7.1) suggests the boundary moves as capabilities improve. The pre-treatment proxy is not a direct LLM-capability benchmark; we report the embedding-answerability treatment as a co-primary check, but a direct pass-rate measure (a local open-weight model over sampled pre-period questions) would be more precise and is a natural next step within the zero-budget rule.

Pre-trends. Exact parallel trends are rejected; we treat the pre-period convergence, placebo sign-reversal, Copilot placebo, and the HonestDiD Δ^{RM} profile (robust CI first contains zero at $\bar{M} = 0.75$; §4.2) as evidence against a simple secular-trend explanation, not proof of parallel trends, and describe the causal reading as moderate rather than strong. A quadratic trend control (in the replication package) absorbs more of the effect mechanically by mimicking the shock shape; we report it as a limit case.

No welfare conclusion. We bound one public-cost channel from above. The private productivity gains documented elsewhere are first-order and plausibly an order of magnitude larger (§9.2), and the long-run model-degradation implications (Shumailov et al., 2024) are unresolved.

Scope of the artefact. The early-warning system is a reproducible scoring-and-alerting script (§9.1); a production dashboard and a usability evaluation with platform moderators would be required to claim a deployed, human-evaluated system, and neither was available for this study. (The 2025 panel extension earlier flagged as future work is now complete for all 100 tags and used in §4.3.)

11 Conclusion

Public Q&A platforms codify routine problem-solving and distribute spillovers across future learners. ChatGPT sharply reduced the marginal cost of obtaining the same service privately,

and the recent literature (Burtch et al., 2024; del Rio-Chanona et al., 2024; Xue et al., 2026) documents the resulting aggregate contraction together with an apparent improvement in surviving question quality—a combination compatible with a benign reading under which only the low-value end is pruned. We test that reading at the within-tag value margin.

Using a triple difference on 8.4 million questions (100 top tags, 261 weeks, seven question types), the central finding rejects the *strict* benign-pruning reading: a single pre-specified shape test returns a negative average displacement across the value distribution (-0.167 , $p < 10^{-4}$) that is roughly constant across the low-to-mid crowd-validated range and fades only at the rare elite tail, so the channel is not confined to score-zero posts (the exclusive bins 0 through 4 are each individually significant and Bonferroni-robust; a conservative Romano–Wolf correction over the nested cumulative thresholds places the gradient at the 10% joint level). The volume baseline is -0.108 ($p = 0.021$), strengthening to -0.140 ($p = 0.005$) once the a-priori boundary category advanced-architecture is excluded, and negative and significant in all 32 cells of a specification curve. Because exact pre-trend tests reject, we read the estimate as a conditional post-ChatGPT association, not a natural experiment: its credibility rests on a sign-reversing placebo distribution, monotone event-study growth, a pre-trend-matched design (-0.129), and a HonestDiD breakdown at $\bar{M} = 0.75$. Re-estimating on the full 100-tag panel extended through 2025 strengthens the effect to -0.172 ($p = 0.002$), deepening to -0.68 with no reversion—most consistent with sustained diffusion rather than a transient episode.

We turn the result into a *value-calibrated early-warning system*: out of sample, an activity-decline signal detects where high-value erosion is largest (AUC 0.98 versus 0.57 for a value-weighted detector) and the value-margin estimates calibrate that alert into the high-value posts at risk, with a persistent composition flag ($\rho = 0.90$) classifying intervention type. We do not document an effect on innovation *outcomes*, contributor entry, or productivity, and we derive no welfare conclusion; the Q&A margin is an innovation *input*, and the private productivity gains of LLM adoption are not estimated. The conceptual implication is that

generative AI changes the *value composition*, not only the quantity, of the public knowledge inputs that fuel open innovation—the dimension that innovation policy and management research should follow. The entire pipeline is reproducible from public, zero-budget data; the full robustness, mechanism, and validation detail is in the replication package.

Data and code availability

All evidence necessary to evaluate the main claims is reported in the article. A public replication package provides the code, data-construction scripts, audit logs, derived tables, and extended robustness outputs needed to reproduce the analysis; it is a research-data and code archive, not journal supplementary material. Raw Stack Overflow content is not redistributed; derived statistics and scripts to regenerate the data from public sources are provided where licensing permits. Replication repository: <https://github.com/hlaverde/so-genai-value-margin>.

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